

Data Representation

TA

Data representation refers to the form in which data is stored, processed, and transmitted

Binary Magnitude

TA

0 represents "Positive"

1 represents "Negative"

Prefixes

TA

Binary Prefixes

- Kibibyte
- Mebibyte
- Gibibyte
- Tebibyte

Decimal Prefixes

- Kilobyte
- Megabyte
- Gigabyte
- Terabyte

1024

OR

1000



Confusion!

Makes Big Difference

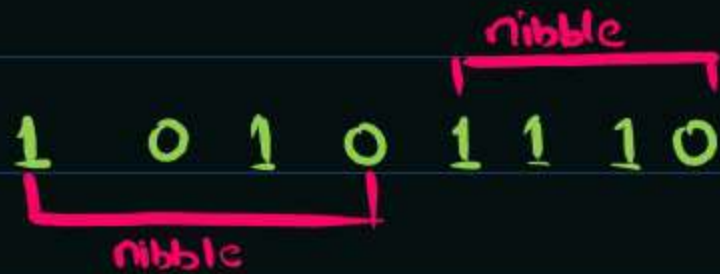
Binary Prefixes



Decimal Prefixes



- Smallest unit is Bit
- One byte contains 8bits
- 4 bits are known as "nibble"
- One Kilobyte contains 1000 bytes
- One Kibibyte contains 1024 bytes
- One Megabyte contains 1024 Kibibytes
- One Megabyte contains 1000 kilobytes



EXAM STYLE QUESTION

- 1 (a) Draw **one** line from each binary value to its equivalent (same) value on the right.

Binary value

8 bits

1000 bytes
8000 bits

1000 kilobytes

1024 mebibytes

8192 bits

1024 bytes

1 byte = 8 bits

1 kibibyte

1 gigabyte

1 byte

1 kilobyte

1 gibibyte

1 megabyte

1 mebibyte

NUMBER SYSTEMS

Binary (base 2)

Denary (base 10)

Hexadecimal (base 16)

Note: Memorized list (128 64 32 16 8 4 2 1)

CASE 1 : DENARY TO BINARY

Question : Convert 65 into 8bit Binary

step 1: Write down the memorized list.

128 64 32 16 8 4 2 1

EXAM STYLE QUESTION

- 2 (a) Convert the following denary integer into 8-bit binary.

55

0	0	1	1	0	1	1	1
128	64	32	16	8	4	2	1

[1]

Case 2 : BINARY To DENARY

Question: Convert 01000001 to Denary

Step 1 : Write the memorized list

0	1	0	0	0	0	0	1
128	64	32	16	8	4	2	1

EXAM STYLE QUESTION

- 2 (a) Convert the following 8-bit binary integer into denary.

01001101
6 5 4 3 2 1
4 2 1

77

[1]

$$64 + 8 + 4 + 1 = 77$$

CASE 3 : BINARY To HEXADECIMAL

value Box

A = 10

B = 11

C = 12

D = 13

E = 14

F = 15

Question : Convert 11011001 into hexadecimal

Step 1 : From right side, divide the binary value into groups of four.

1101 1001

Step 2 : Write the memorized list separately for each group

1101	1001
8 4 2 1	8 4 2 1

Step 3: Add the number with 1 and if it is greater than 9 then use value box

Value Box

A	=	10
B	=	11
C	=	12
D	=	13
E	=	14
F	=	15

1 1 0 1
8 4 2 1

13

1 0 0 1
8 4 2 1

9

D9 → Answer

Exam Style Question

(b) The current contents of a general purpose register (X) are:

X	1	1	1	1	0	0	1	0
	8	4	2	1	8	4	2	1

(i) The contents of X represent an unsigned binary integer.

Convert the value in X into denary.

F2

[1]

CASE 4 : DENARY TO HEXADECIMAL

Question : Convert 195 to Hexadecimal

Step 1 : First convert the denary number into binary

1	1	0	0	0	0	1	1
128	64	32	16	8	4	2	1

Step 2 : Repeat the steps in case 3

1	1	0	0		0	0	1	1
8	4	2	1		8	4	2	1

12

3

(3)  Answer

EXAM STYLE QUESTION

(d) Convert the following hexadecimal number into denary.

$E = 14$

4E

78

[1]

0	1	0	0
8	4	2	1

1	1	1	0
8	4	2	1

0	1	0	0	1	1	1	0
128	64	32	16	8	4	2	1

$$64 + 8 + 4 + 2 = 78$$

BINARY CODED DECIMAL

Uses a 4-Bit code to represent each denary digit

0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001

CASE 1 : DENARY TO BCD

Question : Convert 54 into BCD

Step 1 : Convert each number into BCD format.

5	4
0101	0100

Answer : 01010100

Exam Style Question

(b) (i) Convert the following denary integer into Binary Coded Decimal (BCD).

653

0110 0101 0011

[1]

CASE 2 : BCD To DENARY

Question : Convert 0101 0100 BCD to Denary

Step 1 : Make groups of 4

0101 0100

Step 2: Write 8421 on each group and add the numbers with 1

0101

8421

0100

8421

5

4

Answer = 54

EXAM STYLE QUESTION

(b) Convert the following Binary Coded Decimal (BCD) number into denary.

10000011
8 4 2 1 8 4 2 1

83

[1]

Question: Describe how denary integers larger than 9 can be converted into BCD. Give an example in your answer.

Answer: Each denary digit is written as a 4-bit binary number

Example $46 = 0100\ 0110$

Question: Describe how an 8-bit BCD can be converted into a denary integer. Give an example in your answer.

Answer: Binary Number is split into groups of 4-Bits (starting from the right)
Each group of 4-bits is converted to a denary digit
example 0011 0111 = 37

Question: Identify practical applications where BCD is used.

Answer: Any scenario where a single digit needs to be transmitted / displayed

e.g: Calculator / digital clock

BINARY ADDITION

		S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1
1	1	1	1

Questions :

①

$$\begin{array}{r} 10 \\ 11 \\ \hline 101 \end{array}$$

②

$$\begin{array}{r} 0011 \\ 0011 \\ \hline 110 \end{array}$$

③

$$\begin{array}{r} 01110 \\ 00111 \\ \hline 10101 \end{array}$$

④

$$\begin{array}{r}
 1 \quad 0 \quad 1 \\
 0 \quad 0 \quad 1 \\
 0 \quad 0 \quad 1 \\
 \hline
 1 \quad 0 \quad 0 \quad 1 \quad 0
 \end{array}$$

1 1 0 0
0 1 1 0

⑤

$$\begin{array}{r}
 1 \quad 1 \quad 1 \\
 0 \quad 0 \quad 1 \\
 1 \quad 0 \quad 1 \quad 1 \quad 1 \\
 \hline
 1 \quad 0 \quad 0 \quad 1 \quad 0 \quad 1
 \end{array}$$

1 1 1 0
1 0 1 1 1

⑥

$$\begin{array}{r}
 1 \quad 1 \quad 1 \quad 1 \\
 1 \quad 1 \quad 1 \quad 1 \quad 1 \\
 1 \quad 1 \quad 1 \quad 0 \quad 1 \\
 \hline
 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0
 \end{array}$$

1 1 1 1
1 1 1 0 1

EXAM STYLE QUESTION

- (ii) The amount of green in binary is 00100011. This has the denary number 15 added to it to create a second colour.

Add the denary number 15 to the binary number 00100011 and give your answer in binary.

Perform the addition in binary. Show your working.

[illegible]

Answer (in binary) 00110010

(b) (i) Perform the following binary addition. Show your working.

$$\begin{array}{r} 11110 \\ + 00110111 \\ \hline 11100001 \end{array}$$

[2]

CONVERSION OF NEGATIVE NUMBERS

Question: Convert -125 To Binary Form.

Step 1: Ignore negative sign and convert it to binary

0	1	1	1	1	1	0	1
128	64	32	16	8	4	2	1

0 1 1 1 1 1 0 1

Step 2 : Apply 2's complement

0 1 1 1 1 1 0 1

1 0 0 0 0 0 1 0 $\rightsquigarrow$ 1's complement

0 0 0 0 0 0 0 1

1 0 0 0 0 0 1 1

Answer = 1 0 0 0 0 0 1 1

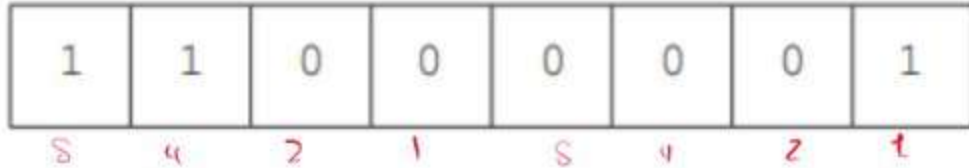
EXAM STYLE QUESTION

(b) Convert the denary number -194 into 12-bit two's complement.

..... [1]

Note : All 2's complement numbers that starts with 1 are negative number

(c) H is a register. The current contents of H are:



The current contents of register H represent an unsigned binary integer.

(i) Convert the value in register H into denary.

.....193.....[1]

(ii) Convert the value in register H into hexadecimal.

.....C1.....[1]

(iii) The current contents of register H represent a two's complement binary integer.

Convert the value in register H into denary.

-63.....[1]

(iv) State why register H does not currently contain a Binary Coded Decimal (BCD).

The first 4 bits would be 12 which is greater than 9
which is not valid for BCD.....[1]

1 1 0 0 0 0 0 1

0 0 1 1 1 1 1 0

Inverse

0 0 1 1 1 1 1 1

128 64 32 16 8 4 2 1

$$32 + 16 + 8 + 4 + 2 + 1$$

$$-63$$

(c) The current contents of a general purpose register (X) are:

X	1	0	1	1	1	0	1	0
	128	64	32	16	8	4	2	1

(i) The contents of X represent an unsigned binary integer.

Convert the value in X into denary.

.....186.....[1]

(ii) The contents of X represent an unsigned binary integer.

Convert the value in X into hexadecimal.

.....BA.....[1]

(iii) The contents of X represent a two's complement binary integer.

Convert the value in X into denary.

.....-70.....[1]

(3c) the current contents of a general purpose register (X) are:

X	1	1	0	0	0	0	0	1
---	---	---	---	---	---	---	---	---

(i) The contents of X represent an unsigned binary integer.

Convert the value in X into denary.

..... [1]

(ii) The contents of X represent an unsigned binary integer.

Convert the value in X into hexadecimal.

..... [1]

(iii) The contents of X represent a two's complement binary integer.

Convert the value in X into denary.

..... [1]

(iii) Express the following integer in two's complement form.

-17

--	--	--	--	--	--	--	--

[1]

BINARY SUBTRACTION

Explanation :

$$40 - 15 = 25$$

OR

$$40 + -15 = 25$$

Note : Apply two's complement and then same working as binary addition.

EXAM STYLE QUESTION

- (iii) Hexadecimal 23 in two's complement representation is 00100011. The denary number 10 needs to be subtracted from this value.

Subtract the denary number 10 from the two's complement representation 00100011.

Give your answer in binary. Show your working.

Working

Converting 10 to -10	Subtracting
00001010 $\rightarrow$ 10	00100011
1110101	+ 11110110
1	
	00011001
1110110	
Answer (in binary) 00011001	

Concept of Overflow

More bits to store but less space

OS = 64 bit

$$\begin{array}{r}
 0 \\
 + 10101010 \\
 \hline
 10110001
 \end{array}$$

The result is 10110001. The leading 1 is circled in blue, and an arrow points from it to the word "OVERFLOW" written in red.

Question : State how an overflow can occur when adding two binary integers

Answer : The result is a larger number than can be stored in the given number of bits .

ASCII

EXTENDED ASCII

UNICODE

BASIC CONCEPT

67

Binary Representation

0 1 0 0 0 0 1 1
128 64 32 16 8 4 2 1

A

Binary Representation

?

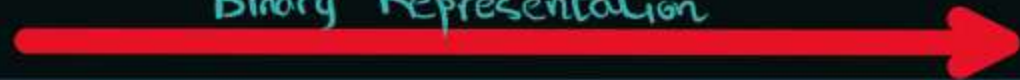
B

Binary Representation

?

[]

Binary Representation



?

Note : the numbers could easily be represented into binary form
but there was no way to represent alphabets, symbols



ASCII Table

Dec	Hex	Oct	Char	Dec	Hex	Oct	Char	Dec	Hex	Oct	Char	Dec	Hex	Oct	Char
0	0	0		32	20	40	[space]	64	40	100	@	96	60	140	`
1	1	1		33	21	41	!	65	41	101	A	97	61	141	a
2	2	2		34	22	42	"	66	42	102	B	98	62	142	b
3	3	3		35	23	43	#	67	43	103	C	99	63	143	c
4	4	4		36	24	44	\$	68	44	104	D	100	64	144	d
5	5	5		37	25	45	%	69	45	105	E	101	65	145	e
6	6	6		38	26	46	&	70	46	106	F	102	66	146	f
7	7	7		39	27	47	'	71	47	107	G	103	67	147	g
8	8	10		40	28	50	(	72	48	110	H	104	68	150	h
9	9	11		41	29	51	)	73	49	111	I	105	69	151	i
10	A	12		42	2A	52	*	74	4A	112	J	106	6A	152	j
11	B	13		43	2B	53	+	75	4B	113	K	107	6B	153	k
12	C	14		44	2C	54	,	76	4C	114	L	108	6C	154	l
13	D	15		45	2D	55	-	77	4D	115	M	109	6D	155	m
14	E	16		46	2E	56	.	78	4E	116	N	110	6E	156	n
15	F	17		47	2F	57	/	79	4F	117	O	111	6F	157	o
16	10	20		48	30	60	0	80	50	120	P	112	70	160	p
17	11	21		49	31	61	1	81	51	121	Q	113	71	161	q
18	12	22		50	32	62	2	82	52	122	R	114	72	162	r
19	13	23		51	33	63	3	83	53	123	S	115	73	163	s
20	14	24		52	34	64	4	84	54	124	T	116	74	164	t
21	15	25		53	35	65	5	85	55	125	U	117	75	165	u
22	16	26		54	36	66	6	86	56	126	V	118	76	166	v
23	17	27		55	37	67	7	87	57	127	W	119	77	167	w
24	18	30		56	38	70	8	88	58	130	X	120	78	170	x
25	19	31		57	39	71	9	89	59	131	Y	121	79	171	y
26	1A	32		58	3A	72	:	90	5A	132	Z	122	7A	172	z
27	1B	33		59	3B	73	;	91	5B	133	[	123	7B	173	{
28	1C	34		60	3C	74	<	92	5C	134	\	124	7C	174	
29	1D	35		61	3D	75	=	93	5D	135	]	125	7D	175	}
30	1E	36		62	3E	76	>	94	5E	136	^	126	7E	176	~
31	1F	37		63	3F	77	?	95	5F	137	_	127	7F	177	

No Need
to Memorize

A = 65
a = 97

ASCII

- 7 bit

1	0	0	0	0	0	1
64	32	16	8	4	2	1

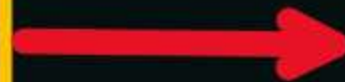
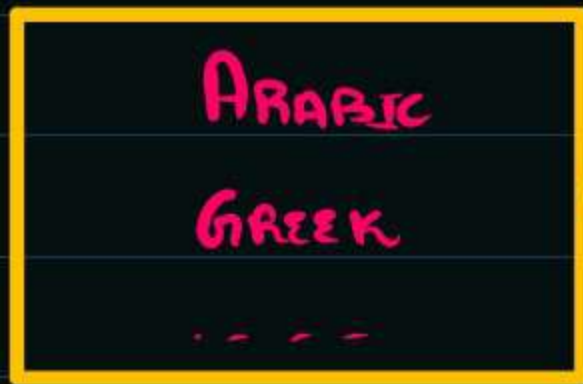
 = A
- 128 characters can be represented
- American Standard Code For information interchange

EXTENDED ASCII

- 8 bits
128 64 32 16 8 4 2 1
- ASCII codes were 7 bits long but then it was extended to have a bit length of 8
- 256 characters can be represented.

UNICODE

PROBLEM



NO REPRESENTATION

• 16, 24, 32 bits

Question: What are the disadvantages of using ASCII code

- Only 128 characters can be represented
- uses value 0 to 127 / One byte
- Many characters used in other languages cannot be represented
- In extended ASCII the characters from 128 to 255 may be coded differently in different system.

Question: How unicode is designed to overcome the disadvantages of ASCII

- Uses 16, 24, 32 bits
- Unicode is designed to be superset of ASCII
- Designed so that most characters in other languages can be represented.

EXAM STYLE QUESTIONS

6 A computer uses the ASCII character set.

- (a) State the number of characters that can be represented by the ASCII character set and the extended ASCII character set.

ASCII 128

Extended ASCII 256

[2]

- (b) Explain how a word such as 'HOUSE' is represented by the ASCII character set.

- Each character has its own unique code
- Each character in the word is replaced by its code
- The codes are stored in the order in the word.

[2]

- (c) Unicode is a different character set.

The Unicode value for the character '1' is denary value 49.

- (i) Write the hexadecimal value for the Unicode character '1'.

31

[1]

- (ii) Write the denary value for the Unicode character '5'.

53

[1]